

Practical Workbook

CT-261

DATABASE MANAGEMENT SYSTEMS



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Practical Workbook

Database Management Systems

(CT - 261)

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LAB#11

PL/SQL Declaring Variables

Q1. Evaluate each of the following declarations. Determine which of them are *not* legal and explain why.

- a) **DECLARE v_id NUMBER(4);**
- b) **DECLARE v_x, v_y, v_z VARCHAR2(10);**
- c) **DECLARE v_birthdate DATE NOT NULL;**
- d) **DECLARE v_in_stock BOOLEAN := 1;**

SOLUTION:

- a) **EXPLANATION:** This is a legal statement. It follows all the rules of PL/SQL Syntax.
- b) **EXPLANATION:** This is not a legal statement, as there are syntax issues. The issue is that you have to separately name the datatype of all variables, and also add a separate semicolon after defining the variables and their datatypes.

CORRECTED STATEMENT:

```
1  DECLARE
2  v_x VARCHAR2(10);
3  v_y VARCHAR2(10);
4  v_z VARCHAR2(10);
```

- c) **EXPLANATION:** This statement is also not legal because we always have to initialize a value when we are using the NOT NULL keyword.

CORRECTED STATEMENT:

```
6  DECLARE
7  v_birthdate DATE NOT NULL:=SYSDATE;
```

- d) **EXPLANATION:** This statement is illegal because the BOOLEAN keyword only works with True/False and not numeric values like 1 or 0.

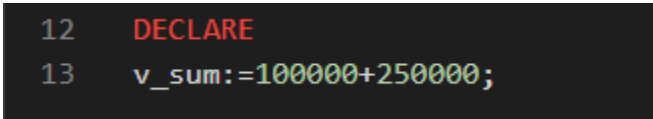
CORRECTED STATEMENT:

```
9  DECLARE
10 v_in_stock BOOLEAN:=TRUE;
```

Q2. In each of the following assignments, indicate whether the statement is valid and what the valid data type of the result will be:

- a) `v_days_to_go := v_due_date - SYSDATE;`
- b) `v_sender := USER || ':' || TO_CHAR(v_dept_no);`
- c) `v_sum := $100,000 + $250,000;`
- d) `v_flag := TRUE;`
- e) `v_n1 := v_n2 > (2 * v_n3);`
- f) `v_value := NULL;`

SOLUTION:

- a) **VALIDITY:** Valid.
RESULT TYPE: NUMBER
EXPLANATION: Subtracting one date from another will represent the difference in days.
- b) **VALIDITY:** Valid
RESULT TYPE: VARCHAR2
EXPLANATION: The concatenation operator merges everything into one single result.
- c) **VALIDITY:** Invalid
RESULT TYPE: NUMBER
EXPLANATION: The assignment is invalid because PL/SQL does not allow the use of symbols and the comma operator, such as the dollar sign or any other type of sign.
CORRECTED STATEMENT:


```
12  DECLARE
13  v_sum:=100000+250000;
```
- d) **VALIDITY:** Valid
RESULT TYPE: BOOLEAN
EXPLANATION: The keyword **TRUE** simply represents the use of **BOOLEAN** datatype.
- e) **VALIDITY:** Valid.
RESULT TYPE: BOOLEAN
EXPLANATION: The assignment is valid, but **v_n1** should be of type **BOOLEAN** otherwise it will be invalid.
- f) **VALIDITY:** Valid
RESULT TYPE: Will inherit variable's original datatype.
EXPLANATION: The assignment is valid because **NULL** keywords can be assigned to variables of all datatype.

Q3. Create a block that declares two variables. Assign the value of these PL/SQL variables to iSQL*Plus host variables and print the results of the PL/SQL variables to the screen. Execute your PL/SQL block. V_CHAR Character (variable length) V_NUM Number. Assign values to these variables as follows:

- V_CHAR The literal '42 is the answer'
- V_NUM The first two characters from V_CHAR

SOLUTION:

```
15 DECLARE
16     V_CHAR VARCHAR2(50);
17     V_NUM  NUMBER;
18 BEGIN
19     V_CHAR:='42 is the answer';
20     V_NUM:=TO_NUMBER(SUBSTR(V_CHAR,1,2));
21
22     DBMS_OUTPUT.PUT_LINE('V_CHAR = ' || V_CHAR);
23     DBMS_OUTPUT.PUT_LINE('V_NUM = ' || V_NUM);
24 END;
25 /
```

OUTPUT:

Results	Explain	Describe	Saved SQL	History
V_CHAR = 42 is the answer V_NUM = 42 Statement processed. 0.01 seconds				